

Name: Student ID:

ECS401: Unit 8 Lab Progress Test

These exercises are to help you judge your progress. It is really important that you make sure you fully understand everything we have covered so far.

1. Complete the dry run table when this code is executed. What is printed?

```
int score1 = 57;           // L1
int score2 = 21;           // L2
score2 = score1;           // L3
score2 = 49;               // L4
System.out.println(score1 + ":" + score2); // L5
```

Line	score1	score2	print

Printed:

2. Complete the dry run table showing what happens when this code is executed. What is printed?

```
int[] score1 = {57, 57};           // L1
int[] score2 = {21, 21};           // L2
score2 = score1;                   // L3
score2[1] = 27;                    // L4
System.out.println(score1[1] + ":" + score2[1]); // L5
```

In this table @1 and @2 represent the addresses of storage space on the heap. When an array is initialised, the data is stored in the storage space at that address and the address is stored in the named variable as shown. The first line completed shows the result of executing the first line of code.

Line	score1	score2	@1		@2		print
			[0]	[1]	[0]	[1]	
L1	@1		57	57			

Printed:

3. Complete the dry run table showing what happens when this code is executed. What is printed?

```

int[] a = {1,2,3,4};           // L1
int[] b = {1,2,3,4};           // L2
int result = 2;                 // L3
if (a==b)                       // L4
    result = 0;                 // L5
else
    result = 1;                 // L6
System.out.println(result);    // L7

```

L	a	b	@1				@2				result	a==b ?	print
			[0]	[1]	[2]	[3]	[1]	[2]	[3]	[4]			

Printed:

4. Consider the following method:

```
public static void wombat(int a, int[] b) // L1
{
    a = 1;                                // L2
    b[0] = 2;                             // L3
    return;                               // L4
}
```

Complete the dry run table. What is printed when the following fragment calls the method:

```
int x = 5;                                // L5
int[] y = {6, 7, 8};                     // L6
wombat(x, y);                             // L7
System.out.println(x + ":" + y[0]);      // L8
```

Line	x	y	@1			wombat(x, y)	print
			[0]	[1]	[2]		

wombat(,)						
Line	a	b	@2			return
			[0]	[1]	[2]	

Printed:

5. Complete the picture showing the way data is stored on the stack and heap after the following sequence of declarations.

```
int x = 5;  
int[] y = {6, 7, 8};  
String animal = "bear";  
char initial = 'p';
```

Draw the String as if fitting in a single cell.

